1. Consider the ring $M_2(\mathbb{R})$.

(a) Take any nonzero 2×2 matrix A. Show that by multiplying A on the left by matrices of the form

$$\begin{bmatrix} 1 & a \\ 0 & 1 \end{bmatrix}, \begin{bmatrix} 1 & 0 \\ b & 1 \end{bmatrix}, \begin{bmatrix} c & 0 \\ 0 & 1 \end{bmatrix}, \begin{bmatrix} 1 & 0 \\ 0 & c \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix},$$

we can do any elementary row operation to A.

(b) State a way of interpreting column operations using matrix multiplication.

(c) Prove that the only ideals in $M_2(\mathbb{R})$ are $\{0\}$ and $M_2(\mathbb{R})$.

(a) Let A be arbitrary matrix in $M_2(\mathbb{R})$

Then $A = \begin{bmatrix} w & x \\ y & z \end{bmatrix}$ for some $w, x, y, z \in \mathbb{R}$

so ① $\begin{bmatrix} 1 & a \\ 0 & 1 \end{bmatrix} A = \begin{bmatrix} w+ay & x+az \\ y & z \end{bmatrix}$

it is equivalent to add some multiple of the second row to the first row.

② $\begin{bmatrix} 1 & 0 \\ b & 1 \end{bmatrix} A = \begin{bmatrix} w & x \\ y+bw & z+bx \end{bmatrix}$

it is equivalent to add some multiple of the first row to the second row

③ $\begin{bmatrix} c & 0 \\ 0 & 1 \end{bmatrix} A = \begin{bmatrix} cw & cx \\ y & z \end{bmatrix}$

it is equivalent to multiplying the first row by some scalar c

④ $\begin{bmatrix} 1 & 0 \\ 0 & c \end{bmatrix} A = \begin{bmatrix} w & x \\ cy & cz \end{bmatrix}$

it is equivalent to multiplying the second row by some scalar c

⑤ $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} A = \begin{bmatrix} y & z \\ w & x \end{bmatrix}$

It is equivalent to swapping the order of the two rows.

By ①,②,③,④,⑤, we have shown that through multiplying A on the left by matrices of the five forms, we can do all five elementary row operations to A (respectively).

(b) Column operations are just multiplying A on the right by the same five matrix in (a)

ex: $A \begin{bmatrix} 1 & a \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} w & x+aw \\ y & z+ay \end{bmatrix}$ is adding some mutiple of the first column to the second.

(c) Pf, We have known that any ring has $\{0\}$ as an ideals.

Now we prove that any ideal of $M_2(\mathbb{R})$,
if is not $\{0\}$, then must be $M_2(\mathbb{R})$ itself.

Let I be an ideal of $M_2(\mathbb{R})$,

Assume $I \neq \{0\}$ $\left(\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}\right)$, so there exists
some othere element $A \neq 0 \in I$

Let $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$

Since $A \neq 0$, at least one of its entry is not 0.

Without Loss of generality, assume $a \neq 0$

So by definition of ideal,

$$\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}\begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} a & b \\ 0 & 0 \end{bmatrix} \in I \quad ⓪$$

$$\Rightarrow \begin{bmatrix} a & b \\ 0 & 0 \end{bmatrix}\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} a & 0 \\ 0 & 0 \end{bmatrix} \in I \quad ①$$

$$\Rightarrow \begin{bmatrix} \frac{1}{a} & 0 \\ 0 & 0 \end{bmatrix}\begin{bmatrix} a & 0 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \in I \quad ②$$

$$\Rightarrow \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \cdot \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \in I, \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \in I$$

$\Rightarrow \underbrace{\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix} \in I}$ ③

$\Rightarrow \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} = \underbrace{\begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \in I}$ ④

By ②④, $\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} + \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} = \underbrace{\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \in I}_{⑤}$ (i.e. $I_2 \in I$)

No matter what entry we assume is not zero, ~~at least~~ we can always get ⑤ since the property of ideal preserves elementary operations by (a)(b), so we can always operate ⓪ to leave only one row and then get $\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$ by elementary operations.

Then we can always get ⑤.

Since ⑤, let $K \in M_2(\mathbb{R})$ be arbitrary,

$$KI = K \in I$$

So $M_2(\mathbb{R}) \subseteq I$.

Since $I \subseteq M_2(\mathbb{R})$, $I = M_2(\mathbb{R})$.

Therefore in conclusion, $I = M_2(\mathbb{R})$ if $I \neq \{0\}$.

Then we have proved the statement.

2. Let $S_{\text{odd}} \subset \mathbb{Q}$ be the subset of rational numbers with odd denominators (when expressed in lowest terms).

(a) Show that S_{odd} is a subring of $\mathbb{Q}$.

(b) Let $I \subseteq S_{\text{odd}}$ be the subset of rational numbers with even numerator (when expressed in lowest terms). Prove that I is an ideal of S_{odd}.

(c) Define a ring homomorphism $\phi\colon S_{\text{odd}} \to \mathbb{Z}_2$. What is the kernel?

(a) ① $1_{\mathbb{Q}} = \frac{1}{1} \in S_{odd}$

$0_{\mathbb{Q}} = \frac{0}{1} \in S_{odd}$

② Let a, b be arbitrary elements of S_{odd},

then $a = \frac{p}{q}$, $b = \frac{m}{n}$ for some $p, q, m, n \in \mathbb{Z}$ by def of rational numbers. Since $a, b \in S_{odd}$, q, n are odd

So $a + b = \dfrac{pn + mq}{qn} \in S_{odd}$ since q, n are odd $\longrightarrow qn$ is odd.

$ab = \dfrac{pm}{qn} \in S_{odd}$ for the same reason

③ Let $a \in S_{odd}$ be arbitrary,

then $a = \frac{p}{q}$ for some $p, q \in \mathbb{Z}$ where q is odd.

so $-a = \frac{-p}{q} \in S_{odd}$

Since ①②③, by theorem 3.2, S_{odd} is a subring of $\mathbb{Q}$

(b) Let a, b be two elements of I then

then $a = \frac{p}{q}$, $b = \frac{m}{n}$ for some $p, q, m, n \in \mathbb{Z}$,

where p, m are even and q, n are odd

So $a + b = \frac{pn + mq}{qn}$.

Since p, m are even, $pn + qm$ is also even

Since q, n are even, qn is also odd.

So $a + b \in I$. (1)

Let x be an arbitrary element of S_{odd}

so $x = \frac{s}{t}$ for some integer s, t where $t \neq 0$, is odd

$\Rightarrow ax = xa = \frac{ps}{tq}$. since t, q are odd,

tq is odd. And since p is even,

ps is even.

$\Rightarrow ax, xa \in I$ (2)

And nonemptiness is guaranteed by $\frac{2}{5} \in I$

So by definition, I is an ideal of S_{odd}.

(c) define $\varphi: S_{odd} \to S$ by mapping all

elements in S_{odd} with even numerator to $[0]_2$

and all elements in S_{odd} with odd numerator to $[1]_2$.

that is : $\boxed{\frac{p}{q} \longmapsto [p]_2}$

① $\varphi(0) = [0]_2$

② $\varphi(1) = \varphi(\frac{1}{1}) = [1]_2$

③ Let a, b be arbitrary element in S_{odd}

then $a = \frac{p}{q}$, $b = \frac{m}{n}$ for some $p, q, m, n \in \mathbb{Z}$, q, n are odd and nonzero

So $\varphi(a)\varphi(b) = [p]_2[m]_2 = [pm]_2 = \varphi(ab)$

$\varphi(a) + \varphi(b) = [p]_2 + [m]_2 = [p+m]_2 = \varphi(a+b)$

Therefore by ①②③, φ is a homomorphism

and $\ker(\varphi) = \{a \in S_{odd} \mid \varphi(a) = [0]_2\}$

$= \{$all elements in S_{odd} with even numerator to $[0]_2\}$ by our definition

$= \{\frac{p}{q} \in S_{odd} \mid p \text{ is even}\}$

3. Let F be a field, and let $f \in \mathbb{F}[x]$. Two polynomials $g, h \in \mathbb{F}[x]$ are **congruent modulo** f if $f|(g-h)$. We write $g \equiv h \mod f$. The set of all polynomials congruent to g modulo f is written $[g]_f$. For this problem, we fix a polynomial $f \in \mathbb{F}[x]$ of degree $d > 0$.

(a) Prove that every congruence class $[g]_f$ contains a *unique* polynomial in the set $S = \{h(x) \in F[x] \; : \; h(x) = 0 \text{ or } \deg h(x) < d\}$.

(b) How many distinct congruence classes are there for $\mathbb{Z}_2[x]$ modulo $x^3 + x$?

(c) How many distinct congruence classes are there for $\mathbb{Z}_3[x]$ modulo $x^2 + x$?

(a) Let $[g]_f$ be an arbitrary congruence class modular f.

Let $k(x)$ be an element in it and fix it.

Guaranteed by the division algorithm,

there exists some $q(x), r(x) \in F[x]$ such that

$$k(x) = q(x) f(x) + r(x), \text{ where } \deg(r(x)) = 0 \text{ or } \deg(r(x)) < \deg f(x) \; \cancel{(d)}$$

so $c(x) - r(x) = q(x) f(x)$

$$f(x) \mid (k(x) - r(x)) \Rightarrow r(x) \in [g]_f$$

So we have proved the existence of such polynomial in S.

No we show uniqueness: fix the $r(x)$,

Let $h(x)$ be an arbitrary element in $[g]_f$

so $r(x) \equiv h(x) \mod f$

then $h(x) = r(x) + m(x)f(x)$ for some $m(x) \in F[x]$

if $m(x) = 0$ then $h(x) = r(x)$, they are the same element.

if $m(x) \neq 0$ then $\deg h(x) \geq \deg f(x)$.

Therefore the $r(x) \in S$ and also in $[g]_f$ is unique.

Here we have finished the proof.

(b)/(c) By (a), every congruent class $[g]_f$ contains a unique polynomial in $S = \{h(x) \mid \deg h(x) = 0$ or $\deg h(x) < \deg f(x)\}$

and by definition, every element of

$\{[h(x)]_f \mid \deg h(x) = 0$ or $\deg h(x) < \deg f(x)\}$

is a unique congruent class modulo f.

So $\{[h(x)]_f \mid \deg h(x) = 0$ or $\deg h(x) < \deg f(x)\}$

is just the set of all congruent classes modulo f.

so we only need to find the number of polynomials that have a smaller degree than $f(x)$

Therefore for $x^3 + x$ in $\mathbb{Z}_2[x]$: $2^3 = 8$ ($\deg = 0, 1, 2$)

for $x^2 + x$ in $\mathbb{Z}_3[x]$: $3^2 = 9$ ($\deg = 0, 1$)

4. What are the subrings of $\mathbb{Q}$? We have $\mathbb{Z}$, $\mathbb{Q}$, and according to the previous problem, the subring S of rational numbers with odd denominators.

(a) Prove that the subset

$$\mathbb{Z}[1/2] := \left\{ \frac{a}{2^m} \middle| a, m \in \mathbb{Z} \text{ and } m \geq 0 \right\} \subset \mathbb{Q}$$

is a subring.

(b) Let $R \subset \mathbb{Q}$ be a subring. Define:

$$\Pi(R) := \left\{ p \text{ a positive prime} \middle| \frac{1}{p} \in R \right\} \subset \mathcal{P} \text{ (the set of positive primes).}$$

Compute $\Pi(\mathbb{Z})$, $\Pi(\mathbb{Q})$, $\Pi(\mathbb{Z}[1/2])$, $\Pi(S_{\text{odd}})$ (no proof needed).

(c) (Tricky!) Given set of the positive prime numbers $\Gamma \subset \mathcal{P}$, define a subring denoted $\mathbb{Z}[1/\Gamma]$ such that $\Pi(\mathbb{Z}[1/\Gamma]) = \Gamma$.

(d) (This is also hard!) Prove that two subrings $R_1, R_2 \subset \mathbb{Q}$ are equal $\iff$ $\Pi(R_1) = \Pi(R_2)$. Conclude, that the subrings of $\mathbb{Q}$ are in bijection with the subsets of the positive prime numbers!

(a) Proof. ① $1_{\mathbb{Q}} = \frac{1}{2^0} \in \mathbb{Z}[\frac{1}{2}]$

$0_{\mathbb{Q}} = \frac{0}{2^0} \in \mathbb{Z}[\frac{1}{2}]$

② Let x, y be arbitrary elements in $\mathbb{Z}[\frac{1}{2}]$

then $x = \frac{a_1}{2^{m_1}}$, $y = \frac{a_2}{2^{m_2}}$ for some $a_1, a_2, m_1, m_2 \in \mathbb{Z}$ with $m_1, m_2 \geq 0$

So $x + y = \frac{a_1 2^{m_2} + a_2 2^{m_1}}{2^{m_1+m_2}}$

since $a_1 2^{m_2}, a_2 2^{m_1} \in \mathbb{Z}$, $m_1 + m_2 \geq 0 \in \mathbb{Z}$,

$x + y \in \mathbb{Z}[\frac{1}{2}]$

$xy = \frac{a_1 a_2}{2^{m_1+m_2}} \in \mathbb{Z}[\frac{1}{2}]$ for the same reason

③ Let z be arbitrary element in $\mathbb{Z}[\frac{1}{2}]$

then $z = \frac{a}{2^m}$ for some $m, a \in \mathbb{Z}$ with $m \geq 0$.

Then $-z = \frac{-a}{2^m} \in \mathbb{Z}[\frac{1}{2}]$

Since ①②③, $\mathbb{Z}[\frac{1}{2}]$ is a subring of $\mathbb{Q}$ by theorem 3.2.

(b) $\Pi(\mathbb{Z}) = \{p \text{ is positive prime} \mid \frac{1}{p} \in \mathbb{Z}\}$

$= \emptyset$

$\Pi(\mathbb{Q}) = \{p \text{ is positive prime} \mid \frac{1}{p} \in \mathbb{Q}\}$

$= \{\text{all positive primes}\}$

$\Pi(\mathbb{Z}[\frac{1}{2}]) = \{p \text{ is positive prime} \mid \frac{1}{p} = \frac{a}{2^m} \text{ for some } a, m \in \mathbb{Z}, m \geq 0\}$

$\{p \text{ is positive prime} \mid p = \frac{2^m}{a} \text{ for some } a, m \in \mathbb{Z}, m \geq 0\}$

$= \{2\}$ since all other primes are not multiples of 2.

$\Pi(S_{odd}) = \{p$ is positive prime $\mid \frac{1}{p}$ is a rational number with odd denominator$\}$

$= \underline{\{\text{all positive primes}\} \setminus \{2\}}$

since every prime is odd except 2

(c)

We want to define $\mathbb{Z}[1/\sigma]$ such that

$\{p$ is positive prime $\mid \frac{1}{p} \in \mathbb{Z}[1/\sigma]\} =$

$\{p \in \mathbb{Z} \mid p \in \sigma\}$

We can define $\mathbb{Z}[1/\sigma]$ as

$\mathbb{Z}[1/\sigma] = \{\frac{a}{p} \mid a \in \mathbb{Z}$, p is the products of some positive primes in $\sigma\}$

$= \{\frac{a}{p_1 p_2 \cdots p_s} \mid a, s \in \mathbb{Z},\ p_1, p_2, \ldots, p_s \in \sigma\}$

We can show that

① $\Pi(\mathbb{Z}[1/\Gamma]) = \Gamma$

② $\mathbb{Z}[1/\Gamma]$ is a subring of $\mathbb{Q}$

①: Let p be an arbitrary element in Γ

take $a=1$, $\frac{1}{p} = \frac{a}{p} \in \mathbb{Z}[1/\Gamma])$

so $p \in \Pi[\mathbb{Z}[1/\Gamma]] \Rightarrow \underline{\Gamma \subseteq \Pi[\mathbb{Z}[1/\Gamma]]}$

Let q be arbitrary element in $\Pi(\mathbb{Z}[1/\Gamma])$

so q is a positive prime and $\frac{1}{q} = \frac{a}{p_1 p_2 \cdots p_s}$

for some $a \in \mathbb{Z}$ and $p_1, p_2, \ldots, p_s \in \Gamma$.

By FTA, $a = q_1 q_2 \cdots q_t$ for some primes $q_1, q_2, \ldots, q_t$

Since $\frac{a}{p_1 p_2 \cdots p_s} = \frac{1}{q}$, $aq = p_1 p_2 \cdots p_s$

$q_1 q_2 \ldots q_t q = p_1 p_2 \cdots p_s$

Since they are all primes, $q_1 q_2 \cdots q_t$ must be the same as some $p_i p_{i+1} \cdots p_{i+t-1}$ after rearrangement

So q is the product of some primes among $p_1, p_2, \ldots, p_s$

Since q is a prime, q is one of the primes among $p_1, p_2, \dots, p_s$.
itself

So $q \in \Gamma \Rightarrow \pi(\mathbb{Z}[1/\gamma]) \subseteq \gamma$.

So $\pi(\mathbb{Z}[1/\gamma]) = \gamma$.

② First, $1 = \frac{5}{5} \in \mathbb{Z}[1/\gamma]$

$0 = \frac{0}{2} \in \mathbb{Z}[1/\gamma]$

Let x, y be arbitrary elements of $\mathbb{Z}[1/\gamma]$.

so $x = \frac{a}{p_1 p_2 \dots p_s}$, $y = \frac{b}{q_1 q_2 \dots q_t}$ for some

$p_1 \dots p_s, q_1 \dots q_t \in \Gamma$ and $a, b \in \mathbb{Z}$

so $x + y = \frac{a(q_1 \dots q_t) + b(p_1 \dots p_s)}{p_1 \dots p_s q_1 \dots q_t} \in \mathbb{Z}[1/\gamma]$

$xy = \frac{ab}{p_1 \dots p_s q_1 \dots q_t} \in \mathbb{Z}[1/\gamma]$

$-x = \frac{-a}{p_1 \dots p_s} \in \mathbb{Z}[1/\gamma]$

So $\mathbb{Z}[1/\gamma]$ is a subring of $\mathbb{Q}$.

(d) $R_1, R_2 \subset \mathbb{Q}$ are subrings

$\pi(R_1) = \{p \text{ is positive prime} \mid \frac{1}{p} \in R_1\}$

$\pi(R_2) = \{p \text{ is positive prime} \mid \frac{1}{p} \in R_2\}$

First, if $R_1 = R_2$ then

$\pi(R_2) = \{p \text{ is positive prime} \mid \frac{1}{p} \in R_1\}$

$= \pi(R_1)$

So to finish this if-and-only-if proof,

we only need to show $\pi(R_2) = \pi(R_1) \Rightarrow R_2 = R_1$

Assume $\pi(R_2) = \pi(R_1)$

Let p be an arbitrary element of R_1

since $R_1 \subset \mathbb{Q}$, $p = \frac{n}{d}$ for some $m, n \in \mathbb{Z}$

where $d \neq 0$, and $(n, d) = 1$.

Since $1 \in R_1$ by def of subring

$1 + 1 + 1 + \ldots + 1 \in R_1$ by def of ring

So recursively we get $\underbrace{\mathbb{Z} \subseteq R_1}$ $\underbrace{(\text{any subring of } \mathbb{Q} \text{ has } \mathbb{Z} \text{ has a subset})}$

since $\underline{\gcd(n,d)=1}$

By Bezout, $\exists\, x, y \in \mathbb{Z}$ s.t.

$$xn + yd = 1$$

$$\Rightarrow \underbrace{x \cdot \frac{n}{d} + y = \frac{1}{d}}$$

since $\frac{n}{d} \in R_1$ and since $x, y \in \mathbb{Z}$, $x, y \in R_1$

so $x \cdot \frac{n}{d} + y = \underbrace{\frac{1}{d} \in R_1}$ by property of ring.

By FTA, $d = p_1 p_2 \ldots p_s$ for some primes $p_1, p_2, \ldots, p_s$

so $\frac{1}{d} = \frac{1}{p_1} \cdot \frac{1}{p_2} \cdot \ldots \cdot \frac{1}{p_s} \in R_1$

Since $p_2, p_3, \ldots, p_s \in R$, by property of ring

$$\frac{1}{d} \cdot (p_2 p_3 \ldots p_s) = \underbrace{\frac{1}{p_1}} \in R_1$$

Similarly we get $\frac{1}{p_2}, \frac{1}{p_3}, \ldots, \frac{1}{p_s} \in R_1$

Since $\underline{\pi(R_1) = \pi(R_2)}$, $\underline{\frac{1}{p_1}, \frac{1}{p_2}, \ldots, \frac{1}{p_s} \in R_2}$

so $\frac{1}{d} = \frac{1}{p_1} \frac{1}{p_2} \cdots \frac{1}{p_s} \in R_2$

Since $\mathbb{Z} \subseteq R_2$ also,

$\frac{n}{d} \in R_2$ by property of ring

So we have proved, $R_1 \subseteq R_2$

And similarly, we can get $R_2 \subseteq R_1$ by exactly the same steps.

So $R_1 = R_2$

Therefore we have finished the proof that

$$\Pi(R_1) = \Pi(R_2) \iff R_1 = R_2$$

Then we can conclude that the subrings of $\mathbb{Q}$ are in bijection with the subsets of the positive prime numbers.